Mid Semester Examination Sept. 2025

Name of the Program: **B-TECH (CSE)** *Semester*: **3^{rd}**

Name of the Course: **Logic Design and Computer Organization**

Course Code: **TCS 308**

Time: 90 Minutes Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions.

(ii) Each question carries 10 marks

Q1	**(10 marks)**	**CO1**
(a)	Simplify the following expression using K-Map method $F(A,B,C,D,E) = \prod M(0, 2, 4, 7, 8, 10, 12, 16, 18, 20, 23, 24, 25, 26, 27, 28)$	
	OR	
(b)	Implement BCD adder and explain with the help of suitable examples.	
Q2	**(10 marks)**	**CO1**
(a)	What are the limitations of normal encoder? Implement priority encoder.	
	OR	
(b)	Implement 3:8 decoder using 1:2 decoder only.	
Q3	**(10 marks)**	**CO1**
(a)	Design a 3 bit code converter which converts binary code to gray code using suitable D-MUX.	
	OR	
(b)	Implement following expression using 4:1 MUX when (a) AB as selected lines. (b) CD as selected lines. $F(A,B,C,D) = \sum m\ (1, 3, 7, 11, 15) + \sum d\ (0, 2,5,8,14)$	
Q4	**(10 marks)**	**CO2**
(a)	Show that the characteristic equation for the complement output of a JK flip-flop is $Q'(t+1) = J'Q' + KQ$	
	OR	
(b)	Convert T flip flop into JK flip-flop.	
Q5	**(10 marks)**	**CO3**
(a)	Draw and explain the circuit of 5 bit SISO right shift register.	
	OR	
(b)	Draw and explain the circuit of MOD 12 up asynchronous counter.	